



Service Tool Instruction

Service Tool Instruction
Number: 3377922-1

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2003

Description

Cylinder Head Coolant Leak Test Kit

Purpose

This document provides information for using cylinder head coolant leak test kit, Part Number 3164917. The kit is used to leak test the cylinder head coolant passages on the Signature, ISX, ISX with CM870, and QSX15 engines.

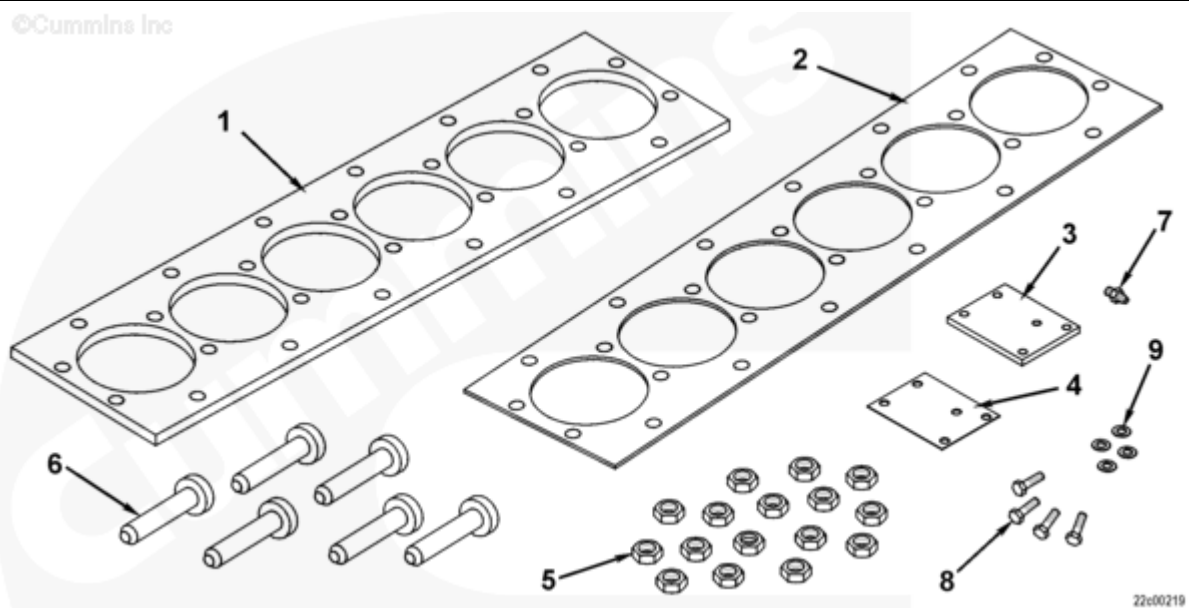


Table 1, Cylinder Head Coolant Leak Test Kit, Part Number 3164917

Item Number	Part Number	Description	Quantity
1	3164918	Plate	1
2	3164919	Gasket	1
3	3164920	Plate	1
4	3164921	Gasket	1
5	3163842	M18 x 1.5 nut	16
6	3164195	Injector sleeve retainer	6
7	68138	Male connector	1

Table 1, Cylinder Head Coolant Leak Test Kit, Part Number 3164917

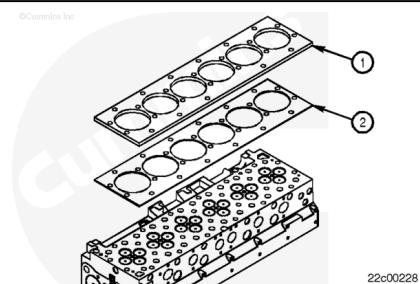
Item Number	Part Number	Description	Quantity
8	3064106	Capscrew	4
9	3009330	Washer	4

Table 2, Items Used with the Cylinder Head Coolant Leak Test Kit, Purchased Separately

Item Number	Part Number	Description	Quantity
Not shown	3164231	Leak test kit	1
Not shown	3162996	Cylinder head lifting bracket	1
Not shown	3375957	Lifting strap	1

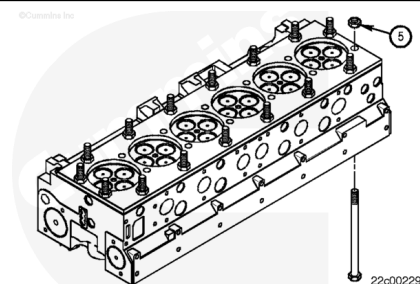
Place the cylinder head on a work bench with the intake side down.

Install the gasket (2) onto the combustion face of the cylinder head. Install the plate (1), with the marking “exhaust” toward the exhaust side of cylinder head, onto the gasket.



Install the cylinder head capscrews and nuts (5), as illustrated. Tighten the capscrews.

Torque Value: 68 n•m [50 ft-lb]

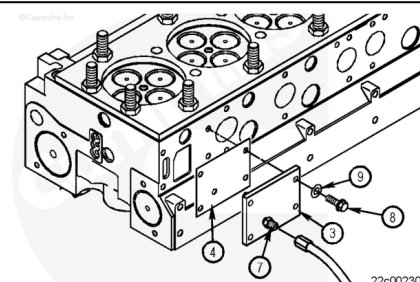


Note : If the male connector (7) is not installed in the plate (3), install it now using thread sealant.

Install the gasket (4) and plate (3) onto the cylinder head thermostat location. Install the four capscrews (8) and washers (9). Tighten the capscrews

Torque Value: 27 n•m [20 ft-lb]

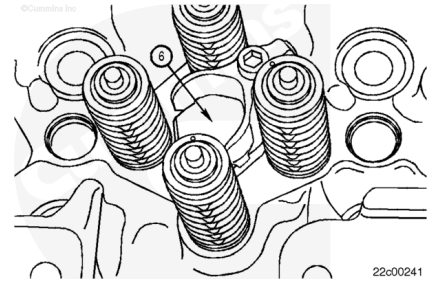
Attach the leak test kit, Part Number 3164231, or equivalent, to the male connector (7).



Note : Make certain the injector sleeve retainers are centered in the injector bore when the clamps are tightened.

Install the six injector sleeve retainers (6) into the cylinder head injector bores. Install the injector hold down clamps and capscrews. Tighten the capscrews.

Torque Value: 41 n·m [30 ft-lb]

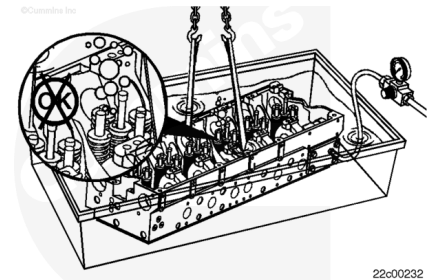


⚠ WARNING ⚠

This assembly weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift this assembly.

Leak Test with Dunk Tank

Use cylinder head lifting bracket, Part Number 3162996, or lifting strap, Part Number 3375957, and a hoist to lower the cylinder head into a dunk tank. Apply 207 kPa [30 psi] of air pressure. Allow several minutes for stabilization. If bubbles are present, check to make sure the plates, gaskets, and male connector are **not** leaking. If bubbles persist and are **not** attributed to the above service tools, the cylinder head **must** be replaced.

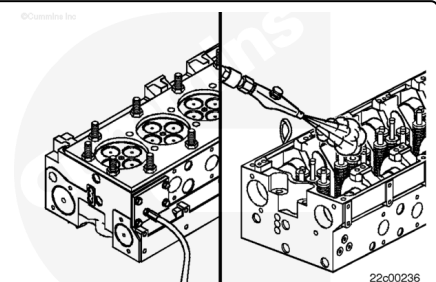


Leak Test without Dunk Tank

Apply 207 kPa [30 psi] of air pressure. Spray a mild soapy water solution around the six injector bores, cup plugs, exhaust ports, and entire combustion face.

Check for the presence of bubbles. If bubbles are present, check to make sure the gaskets, plates, and male connector are **not** leaking.

If bubbles persist and are **not** attributed to the above service tools, the cylinder head **must** be replaced.

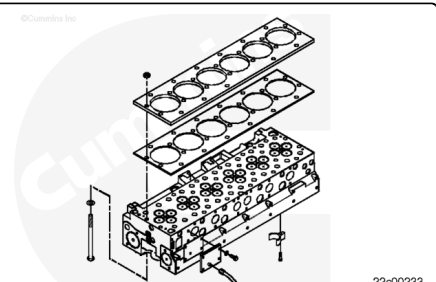


Remove the cylinder head capscrews and nuts.

Remove the plate and gasket.

Remove the air supply hose, capscrews, washers, plate, and gasket.

Remove the six injector sleeve retainers, capscrews and hold down clamps, from the cylinder head injector bore.



Last Modified: 02-Apr-2003
